

Prajwal Narendra Tidke

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Professional Summary

Data Engineer with 5+ years building high-scale batch and streaming pipelines processing multi-terabyte daily volumes across manufacturing, financial services, and healthcare. Proficient in optimizing lakehouse architectures on AWS, Databricks, and Snowflake, with recent work building production data pipelines for RAG and LLM-powered applications.

Education

San Diego State University <i>MS in Big Data Analytics (Academic Excellence Award, CGPA: 4.0/4.0)</i>	August 2024 – May 2026 San Diego, CA
Indian Institute of Information Technology <i>BTech in Computer Science and Engineering</i>	August 2016 – May 2020 Jabalpur, India

Technical Skills

Languages: Python, SQL, Pandas, PySpark, Scala, Java
Pipelines & Orchestration: Airflow, Spark, Kafka, Flink, Faust, Hive, Hadoop, dbt, SCD (Type 1 & 2), Data Contracts, Data Quality
Data Modeling & Storage: Medallion (Bronze/Silver/Gold), Dimensional Modeling, ETL/ELT, Delta Lake, Apache Iceberg, Parquet, MinIO
Warehouses, Databases & BI: Snowflake, Databricks, BigQuery, SQL Server, Oracle, PostgreSQL, DuckDB, Supabase, Power BI, QuickSight
Cloud & Infrastructure: AWS (S3, Glue, EMR, Athena, Lambda, RDS, SQS/SNS, MWAA), Azure, GCP, Terraform, Docker, Kubernetes
AI Systems: RAG, LLMs, LangChain, ChromaDB, Azure OpenAI, Prompt Engineering, RAGAS, LLM-as-a-Judge, MCP, Claude Code
MLOps & Serving: FastAPI, SageMaker, GitHub Actions, CI/CD, Grafana, CloudWatch, Weights & Biases

Experience

San Diego State University - LINC Lab <i>Graduate AI Data Engineer</i>	San Diego, USA February 2025 – May 2026
- Owned an ingestion pipeline parsing 1.5K+ multi-format clinical documents into ChromaDB via Google LangExtract, automating metadata extraction and removing 15 hours/week of manual annotation. - Designed a metadata-tagged chunking and semantic-filtering retrieval layer over ChromaDB and Azure OpenAI, cutting p95 query latency by 55% (3.8s to 1.7s) for KALM. (Link to KALM mental health companion) - Deployed a RAGAS and LLM-as-a-judge evaluation harness gating answer quality at a 0.96 faithfulness threshold across a 200-query clinical test set, with HIPAA-aligned PII/PHI redaction at ingest. - Designed and built a 5-stage deterministic state machine for KALM's crisis-response routing, with defined fallback states for high-risk inputs.	

TresVista Analytics <i>Senior Data Engineer</i>	Pune, India October 2023 – August 2024
- Designed ELT pipelines on Databricks Medallion Architecture orchestrating portfolio analytics for a \$550B+ AUM fund-of-funds client, improving pipeline processing reliability by 30%. - Architected an event-driven ingestion system (AWS SQS/SNS to S3) processing 1.2 TB/day, orchestrated with Airflow on AWS MWAA, and tuned high-cardinality tables with Delta Lake Z-Ordering to cut compute cost \$42K/year and median query latency by 38%. - Integrated 7 market and alternative-data feeds (S&P Global, Visible Alpha, Placer.ai, Similarweb) via a custom API connector, enforcing data contracts in PyTest CI/CD gates to cut weekly manual analysis by 45% at 100% SLA over 6 months. - Automated onboarding infrastructure with Terraform, codifying reproducible cloud environments so new engineers could provision their own sandboxes instead of waiting on manual setup.	

LTIMindtree <i>Senior Data Engineer</i>	Pune, India July 2021 – October 2023
- Led the migration of Carrier Global's legacy on-premise data infrastructure (SQL Server, Oracle) to AWS Glue and Snowflake with SCD Type 2 history tracking, securing \$650K/year in operating savings and improving pipeline data freshness SLAs to 99.9%. - Built curated reporting datasets and refresh pipelines feeding Power BI (NA) and AWS QuickSight (EU) across SQL Server, Oracle, and Snowflake sources, standardizing data definitions and metrics consumed by C-suite across both regions. - Directed technical planning, code reviews, and Agile delivery across 5+ manufacturing and supply chain engagements spanning Carrier's North America operations, consistently meeting SLA targets.	

Data Engineer	July 2020 – July 2021
- Overhauled legacy batch ETL workflows, tuning Spark and Snowflake query performance via predicate pushdown and indexing strategies to accelerate monthly regulatory reporting timelines by 80%. - Built a schema-validation framework across 7+ downstream pipelines, catching schema drift and cutting production data errors by 63%.	

Projects

Risk Pulse – Real-Time Portfolio Risk Intelligence Platform	GitHub Link
- Developed a real-time Kafka pipeline ingesting 8 equity symbols, computing 5-min VWAP, volatility, and drawdown alerts into a date-partitioned Parquet lake on embedded DuckDB; cut warm p50 from 11ms to 0.5ms by removing a cross-schema resolution cost. - Modeled dbt marts with enforced data contracts (typed schemas, primary-key and not-null constraints), 10 data-quality tests, and column-level lineage, failing builds on schema drift before the serving layer. - Built an Apache Iceberg table on MinIO as a snapshot of the Parquet lake and benchmarked Parquet vs Iceberg read latency; layered a ChromaDB and Azure OpenAI RAG service over the marts for natural-language risk queries.	

Awards & Certifications

Microsoft Certified - Azure Databricks Data Engineer Associate (DP-750)	Issued 2026
Microsoft Certified - Azure AI Cloud Developer Associate (AI-200)	Issued 2026
LTIMindtree - LTI GoMX Business Unit of the Year	Awarded 2022